

Takeaway: Coupled Axial Green Solvers

GreenNet learns line-wise Green kernels; CouplingNet learns the directional source split.

A 2D elliptic solve is represented as axial Green inversions plus a learned, balance-preserving source decomposition.

AXIAL GREEN KERNELS

Normalize axial intervals and learn line-wise inverse kernels.

ANALYTIC STRUCTURE

Encode singular Green behavior before neural correction.

SOURCE SPLIT

Predict ϕ/ψ so line-wise inverses form a 2D solution.

ENERGY BOUND

Tie split consistency to final solution error under assumptions.

Axial Green operators provide the line-wise inverses, and CouplingNet supplies the missing source coupling.